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RESEARCH ARTICLE

# An Adaptive Feature Selection Algorithm for Student Performance Prediction

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**ABSTRACT** Educational Data Mining (EDM) is used to ameliorate the teaching and learning process by analyzing and classifying data that can be applied to predict the students' academic performance, and students' dropout rate, as well as instructors' performance. The prediction of student performance is complicated by the vast and diverse range of variables from academic records to behavioral and health metrics. In this paper, we have introduced a new Adaptive Feature Selection Algorithm (AFSA) by amalgamating an ensemble approach for initial feature ranking with normalized mean ranking from five distinct methods to enhance robustness. The proposed method iteratively selects the best features by adjusting its threshold based on each feature's rank to ensure significant contributions to model accuracy and also effectively reduces dataset complexity. We have tested the performance of the proposed feature selection algorithm using five machine learning classifiers: Logistic Regression (LR), K-Nearest Neighbour (KNN), Support Vector Machine (SVM), Naïve Bayes (NB) classifier, and Decision Tree (DT) classifier on four student performance datasets. The experimental results highlight the proposed method significantly decreases feature count by an average feature reduction factor of 5.7, significantly streamlining datasets while maintaining competitive cross-validation accuracy, marking it as a valuable tool in the field of educational data analytics.

**INDEX TERMS** Classification, educational data mining, feature selection, machine learning.

## I. INTRODUCTION

Educational Data Mining (EDM) is an emerging field of machine learning (ML), the practice of applying data mining (DM) techniques in education, has gained significant momentum in recent years [1], [2], [3]. This burgeoning field leverages data mining, machine learning, and statistical methods to analyze educational Big Data, with a particular emphasis on predicting students' performance. It is crucial to understand that evaluating student performance involves more than just analyzing grades; it requires a comprehensive assessment that takes into account factors like course difficulty and individual grading standards [3]. EDM's ability to offer predictive insights is invaluable for educational institutions, enabling them to optimize resources and provide targeted support to students [2], [3].

The advent of the digital age in the 21st century has precipitated a data revolution, profoundly transforming numerous sectors with the emergence of Big Data. This revolution is particularly pronounced in the educational space, where an extensive variety of data is continuously generated through platforms such as student information systems, learning management systems, and massive open online courses (MOOCs) [1], [2]. This influx of educational data brings about a new era of possibilities, offering deep insights that can fundamentally reshape the educational landscape and enhance teaching approaches [3]. As the digital footprint in education grows exponentially, the data collection space spans an extensive spectrum, encompassing academic achievements, demographic information, behavioral indicators, and health-related metrics. The challenge, however, lies

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in managing the vast and varied nature of this data. In the realm of predictive analytics, especially concerning student performance, the plethora of features present significant hurdles, including computational demands, the risk of overfitting, and diminished clarity in model interpretation [4], [5]. Therefore, the development of a sophisticated system capable of intelligently sifting through and prioritizing key features for effective predictive analysis becomes paramount [6].

To address this critical need, this study introduced an Adaptive Feature Selection Algorithm (AFSA) that synergized the strengths of multiple feature ranking methodologies which significantly extends our previous work [7], [8] on student performance prediction. It is designed to identify and focus on the most influential features within the intricate datasets typical in education. Its unique capability lies in effectively simplifying these datasets, thereby enhancing the efficiency and interpretability of predictive models. This approach not only reduces computational complexity but also aids in pinpointing the essential elements that drive educational outcomes, thereby enabling informed decision-making and impactful instructional strategies. In this paper, we have delved into the complexities of the proposed feature selection algorithm, elucidating its theoretical underpinnings and practical implementation in the diverse world of educational Big Data. The goal is to highlight how the proposed feature selection algorithm can transform the prediction and enhancement of student performance by leveraging the insights derived from EDM. We aim to demonstrate the proposed feature selection algorithm's effectiveness in tackling the challenges posed by the vastness and complexity of educational data, thereby establishing it as a crucial tool for educators and researchers in the digital era.

### A. RESEARCH QUESTION AND MOTIVATION

The primary motivation for this study is to address the challenge of feature overload in educational data analytics. The research question we aim to answer is: How can we effectively reduce the complexity of educational datasets while ensuring that the predictive models retain their accuracy and efficiency? This study seeks to fill the gap in existing research by providing a feature selection algorithm that balances dataset simplification with the preservation of essential predictive capabilities.

### B. GOALS AND OBJECTIVES

Guided by this research question, our objectives are twofold. First, we aim to develop an intelligent feature selection algorithm that can efficiently identify and prioritize the most impactful features in extensive educational datasets. Second, we intend to assess the impact of our proposed algorithm on various performance metrics, including accuracy, precision, recall, F1 score, feature reduction factor, and training time. By focusing on these goals, this study contributes to the field of EDM by offering a tool that not only tackles the complexities inherent in large datasets but also enhances the practical application of predictive analytics in education.

The paper is structured as follows: Section II presents a review of the relevant literature on EDM analysis and the usage of feature selection techniques on educational data. Section III presents an overview of feature selection techniques. Section IV describes the methodology and theoretical framework of the proposed Adaptive Feature Selection Algorithm (AFSA) along with other selected feature selection algorithms. Section V outlines the experimental setup and discusses the results and evaluation metrics. The paper concludes in Section VI, where we summarise our findings, acknowledge limitations, and suggest future research directions.

## II. LITERATURE REVIEW

In recent years, the field of education has experienced a surge in interest regarding the utilization of data mining techniques, particularly within the framework of Educational Data Mining (EDM). EDM represents the convergence of data mining technology and educational data, with the primary goal of gaining deeper insights into students' learning processes and enhancing the overall educational experience [9], [10]. Educational practitioners and researchers have consistently emphasized the pressing need for meaningful EDM tools that guide decision-making and provide valuable insights [9], [10]. One of the notable areas of research within EDM is the application of data mining algorithms for the analysis and prediction of academic performance. This research landscape has been summarised in Table 1, highlighting the diverse methods and datasets used by researchers to achieve various objectives.

Feature selection is a pivotal component within the realm of Educational Data Mining, playing a crucial role in enhancing the refinement and accuracy of predictive models, as noted by Estrera and Ramaswami [21], [22]. In the pursuit of identifying key features closely linked to student performance, various techniques, such as Chi-square Statistics and Information Gain, have been widely employed, as highlighted by Febro [23]. Nevertheless, existing research often adopts a myopic approach, focusing primarily on individual algorithms or isolated feature selection techniques, thereby contributing to a fragmented understanding of the domain, as emphasized by Zaffar et al. [24]. This underscores the pressing need for comprehensive methodologies that seamlessly integrate feature selection with predictive modeling. Table 2 provides a succinct summary of diverse feature selection methods employed in EDM research, including manual selection [25], [26], [27], [28], [29], [30], filter-based techniques such as correlation and information gain [31], [32], [33], [34], [35], [36], [37], and wrapper methods such as genetic algorithms and Principal Component Analysis (PCA) [38], [39], [40], [41], [42], [43]. These diverse approaches collectively contribute to the evolving landscape of feature

**TABLE 1. Summary of research studies of educational data mining.**

| Year | Method | Dataset | Objective | Reference |
|---|---|---|---|---|
| 2020 | Naive Bayes, J48 | Educational Benchmark Dataset | Data Analysis | Karthikeyan et al. [11] |
| 2020 | Association Rule Mining, Principal Component Analysis | Real Academic Data | Data Analysis | Crivei et al. [12] |
| 2020 | Genetic Algorithm | Kaggle Repository Data | Prediction | Farissi et al. [13] |
| 2021 | Text Mining, $k$-Nearest Neighbors | Student Teaching Evaluation Data | Data Analysis | Okoye et al. [14] |
| 2021 | Naive Bayes, Random Forest | Benchmark Student Data | Data Analysis | Kumar et al. [15] |
| 2021 | Harris Hawks Optimization | UCI Dataset | Prediction | Turabieh et al. [16] |
| 2021 | Deep Neural Network | Public 4-Year University Data | Prediction | Nabil et al. [17] |
| 2022 | Improved Evolutionary Algorithm with Neuro-Fuzzy Classification, Chaotic Whale Optimization | Benchmark Student Performance Data | Data Analysis | Duhayyim et al. [18] |
| 2022 | Deep Cognitive Diagnosis Framework, Improved K-means, CNN | Real-World Data | Prediction | Gao et al. [19] |
| 2022 | Improved K-means, CNN | University Dataset | Data Analysis and Prediction | Feng et al. [20] |

**TABLE 2. Summary of feature selection methods in EDM.**

| Method | Type | Study |
|---|---|---|
| Manual selection | | [25], [26], [27], [28], [29], [30] |
| Filter | Correlation | [31], [32], [33], [34], [35], [36], [37] |
| | Information Gain | [44], [45] |
| | Classifier | [38], [39] |
| Wrapper | Genetic algorithm | [40], [41], [42] |
| Dimensionality Reduction | PCA | [43] |

selection in EDM, paving the way for more robust predictive models.

While the current landscape of educational data mining underscores the importance of integrating feature selection with prediction modeling, a noticeable research gap exists. Many studies have explored either feature selection or predictive modeling in isolation, leaving a dearth of comprehensive approaches that bridge these two aspects cohesively [46]. This research gap highlights the imperative need for innovative methodologies, such as the Adaptive Feature Selection Algorithm (AFSA), which holds the promise of amalgamating feature selection with predictive modeling. AFSA offers a potential solution to enhance the robustness of student performance predictions, addressing a critical need in the field of EDM. In summary, the synergy between data mining techniques and educational data in the context of EDM has gained substantial attention, particularly in the analysis and prediction of academic performance. Feature selection has emerged as a pivotal aspect within this domain, as evidenced by the studies summarised in Table 1. However, existing research often lacks a cohesive integration of feature selection with predictive modeling. The introduction of the Adaptive Feature Selection Algorithm represents a promising step toward bridging this gap and enhancing the accuracy and interpretability of student performance predictions in the educational landscape.

## III. FEATURE SELECTION TECHNIQUES

In various fields such as machine learning and data analysis, the process of selecting the most relevant features from a high-dimensional dataset plays a critical role in building accurate predictive models. This section explores different feature selection techniques, highlighting their strengths, weaknesses, and applications, drawing insights from several research papers for reference.

### A. FILTER METHODS

Filter methods represent a category of feature selection techniques that assess the relevance of each feature individually. Notable examples include statistical tests like chi-squared tests, mutual information, and correlation-based feature selection (CFS) [47], [48]. These methods generate a ranked list of features based on their individual significance, making them particularly suitable for datasets with a high number of features. However, they may not adequately capture complex interdependencies between features and are less effective at uncovering intricate feature relationships in the data. This limitation highlights the need for more advanced techniques, such as wrapper and embedded methods, that consider feature interactions and offer greater predictive power by integrating with the classifier algorithm. The choice of an appropriate significance threshold for filter methods can be crucial, and techniques like controlling for family-wise error rate (FWER) or false discovery rate (FDR) through methods like Bonferroni correction or Benjamini-Hochberg procedure are often employed to address this challenge. Additionally, in machine learning applications, the selection of the optimum threshold can be treated as a hyper-parameter and determined through cross-validation as part of the model selection process.

### B. WRAPPER METHODS

Wrapper methods utilize the performance of a chosen classifier as a metric to select the best feature subset. These methods consider feature interactions and redundancies and aim to identify the best-performing set of features for a given classifier [49], [50]. They are computationally heavier than filter methods and might lead to overfitting in some cases due to the choice of classifier [51]. Unlike filter methods that produce ranked feature lists, wrapper methods generate a "best" feature subset as the output. This means users are spared the task of determining an optimal threshold or the number of selected features, as the output is already a feature subset. However, this approach has its drawbacks, as it does not readily reveal the relative importance of individual features within the selected set. Additionally, wrapper methods are dependent on the chosen classifier, so the selected features may not remain optimal when a different classifier is employed. This lack of generalisability can lead to issues when applying the model to external datasets. Despite these limitations, wrapper methods have been shown to result in superior performance compared to filter methods, making them a valuable tool for feature selection in machine learning tasks.

### C. EMBEDDED METHODS

Embedded methods integrate feature selection within the classifier algorithm itself during training [47]. Examples include decision tree-based algorithms (e.g., decision tree, random forest, gradient boosting) and regularisation models like LASSO or elastic net. These methods strike a balance between filter and wrapper methods, offering computational efficiency compared to wrapper methods while incorporating the classifier's bias into feature selection, which can improve classifier performance [52]. Embedded methods can capture feature interactions, with some decision tree-based algorithms even considering higher-order interactions. However, they typically require careful handling of highly dimensional datasets due to limitations in their ability to detect feature interactions as the number of features increases. Unlike some multivariate filters, decision tree-based algorithms like random forest do not automatically eliminate redundant features, which can impact their performance. To address this, hybrid methods that combine feature selection and random forest or other algorithms have been proposed. Penalized methods like LASSO, on the other hand, can discard redundant features but require explicit inclusion of interaction terms for feature interactions, which can be computationally prohibitive in highly dimensional data settings. Two-stage or hybrid strategies have been suggested to reduce search spaces in such cases.

### D. HYBRID METHODS

Hybrid methods seamlessly blend various feature selection strategies to capitalize on their respective advantages. An illustrative example involves initiating the process with univariate filter methods, which efficiently reduce the feature set size. This initial reduction curtails the computational load for subsequent wrapper or embedded methods, which are more computationally intensive but can capture intricate feature dependencies and interactions [53], [54]. These hybrid approaches strike a balance between computational complexity and performance. They offer a middle-ground solution between simple filter methods and the more computationally demanding wrapper and embedded methods. Notably, they tend to yield superior performance compared to standalone filter methods, while being less computationally burdensome than pure wrapper methods. For instance, Alzubi et al. [55] demonstrated the effectiveness of a hybrid CMIM + RFE-SVM strategy in classifying patients with various conditions, underscoring the superiority of this combined approach. Moreover, hybrid methods, while advantageous, have their limitations. They may inadvertently overlook relevant interacting features with no discernible individual effects, especially if these interactions are exclusively complex. This potential drawback arises because most filter methods are ill-equipped to model feature-feature interactions. However, using filter algorithms that can account for feature interactions can help mitigate this issue.

### E. OTHER FEATURE SELECTION METHODS

We have discussed the feature selection methods that closely relate to our approach, but there are other specialized approaches for feature selection. Integrative methods narrow the feature search by incorporating external knowledge, reducing computational complexity while depending on the completeness of external data sources [56], [57]. Ensemble methods enhance the selection process by amalgamating outputs from multiple algorithms, thereby leveraging their diversity for improved stability and performance [58]. Exhaustive searches, critical for identifying significant higher-order feature interactions, often employ hybrid approaches to manage computational demands effectively [59], [60], [61], [62]. Each method's choice depends on specific dataset characteristics and research goals, often necessitating a combination of various strategies to harness their collective strengths.

## IV. METHODOLOGY

This research delved into investigating the impact of feature selection techniques on student performance prediction using machine learning, with a focus on incorporating insights from educational data mining (EDM). Figure 1 encapsulates the essence of educational data mining, portraying a cyclic process where students actively interact, use, participate, and communicate within educational systems, generating valuable student data. This data, in turn, undergoes mining techniques, providing targeted recommendations to students while simultaneously revealing discovered knowledge to educators. This dual functionality forms a continuous loop. This research investigated the impact of feature selection techniques on student performance prediction using machine learning. The methodology involved data cleaning, exploratory data analysis, preprocessing, feature selection, and evaluation. Multiple machine learning algorithms were employed and evaluated using cross-validation. The experimental results and findings are discussed in detail. A visual representation of the methodology is provided in Figure 2, encapsulating the sequential flow from data cleaning to the evaluation of machine learning algorithms. The incorporation of EDM principles not only enhances the predictive modeling aspect but also aligns the study with the broader context of leveraging mined educational data to offer targeted insights for both students and educators, contributing to the continuous improvement of educational systems.

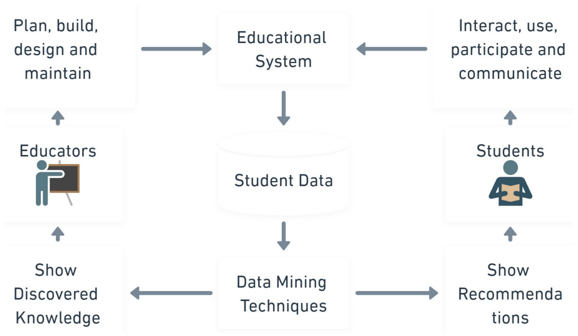



**FIGURE 1.** Different interconnected components of EDM.

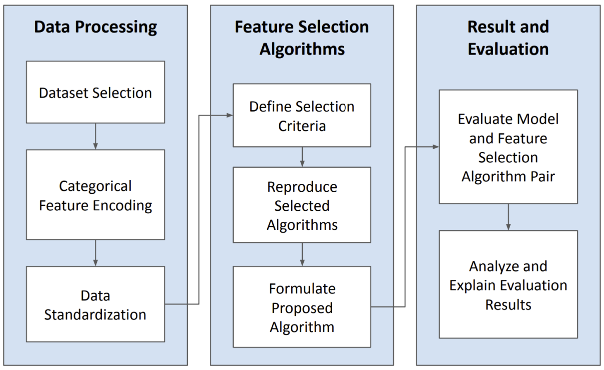



**FIGURE 2.** Block diagram of methodology.

## A. DATASET DESCRIPTION

The research incorporates a comparative analysis of four distinct educational datasets, each offering unique insights into student performance and behavior. The first dataset, xAPI-Educational Mining Dataset (xAPI), was procured from Kaggle in 2016 [44]. Comprising 480 instances and 16 attributes, xAPI includes demographic, academic, and behavioral features primarily tailored for classification tasks. The second dataset, Secondary Student Performance Dataset (SSP), was retrieved from the UCI ML Archive in 2014 [63]. With 1044 instances and 33 attributes, SSP encompasses a diverse range of features, including demographic, academic, behavioral, and health-related attributes. This dataset is versatile, and suitable for both classification and regression tasks. The third dataset, Higher Education Student Performance Dataset (HESP), was sourced from Kaggle in 2022 [64]. Comprising 145 instances and 31 attributes, HESP covers demographic, academic, and behavioral features, making it applicable for classification and regression tasks. The fourth dataset, Student Performance Prediction - Western-OC2-Lab (WOC2), was obtained from GitHub in 2020 [65]. With 486 instances and 9 attributes focusing on academic features, WOC2 is specifically designed for classification tasks. Table 3 provides a comprehensive overview of the key characteristics of these datasets, including the year of acquisition, source, number of instances, number of attributes, types of features, and associated tasks. This comparison

**TABLE 3.** Details comparison of the dataset used in this study.

| Dataset | Year | Source | Instance | Attributes | Feature | Associated Tasks |
|---|---|---|---|---|---|---|
| XAPI [44] | 2016 | Kaggle | 480 | 16 | Demographic, Academic, Behavioral | Classification |
| SSP [63] | 2014 | UCI ML Archive | 1044 | 33 | Demographic, Academic, Behavioral, Health | Classification, Regression |
| HESP [64] | 2022 | Kaggle | 145 | 31 | Demographic, Academic, Behavioral | Classification, Regression |
| WOC2 [65] | 2020 | GitHub | 486 | 9 | Academic | Classification |

aids researchers in understanding the diverse strengths and potential applications of each dataset in educational research and predictive modeling.

## B. EXPLORATORY DATA ANALYSIS

The foundational step in this study involved a comprehensive examination of the dataset through a meticulous process of Exploratory Data Analysis (EDA). This phase aimed not only to ensure the integrity and reliability of the dataset but also to extract meaningful insights that would guide subsequent decisions in data preprocessing, feature engineering, and model selection. The EDA process comprised a multifaceted exploration of the dataset's characteristics, shedding light on its intricacies and patterns.

### 1) DATA CLEANING PROCEDURES

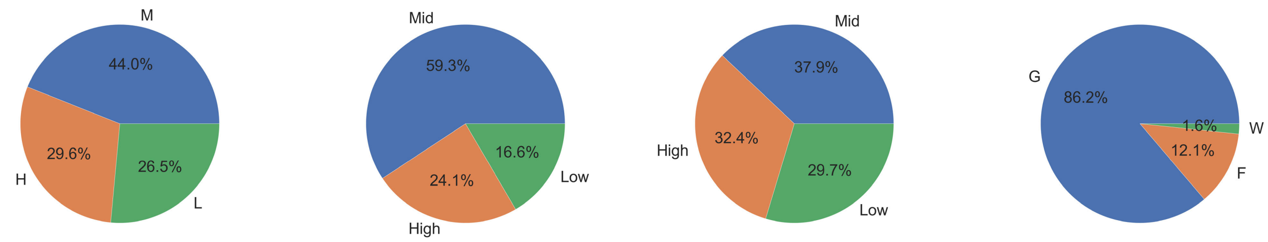
Rigorous data cleaning procedures were implemented to rectify potential discrepancies and enhance the dataset's quality. Missing values were systematically addressed through imputation or removal, ensuring that the dataset was free from gaps that could compromise subsequent analyses. Duplicate entries were identified and removed to maintain data integrity, while columns with constant values were eliminated as they did not contribute meaningful information for model training.

### 2) EXPLORATION OF DATASET CHARACTERISTICS

The heart of the EDA involved a deep dive into the dataset's characteristics. Descriptive statistics were calculated to provide an overview of the dataset. From the pie charts in Figure 3 that shows target column distribution of all four datasets, it can be observed that XAPI, SSP and HESP data distribution is fairly balanced whereas the WCO2 data distribution is imbalanced. This observation has been addressed in the data preprocessing section where state-of-the-art data balancing methods have been used to counteract the issue.

### 3) DATA VISUALISATION

The visualization of data played a crucial role in gaining intuitive insights into complex relationships within the dataset. Correlation heatmaps and class distribution pie charts were employed to visualize the interactions between different




(a) XAPI Dataset (b) SSP dataset (c) HESP dataset (d) WOC2 dataset

**FIGURE 3.** Target column data distribution of four datasets.

features, providing a visual narrative of potential dependencies and correlations. This visual exploration facilitated the identification of potential patterns and outliers, guiding subsequent decisions in the modeling process.

A correlation matrix is a tabular representation that displays the correlation coefficients between multiple variables, where each cell in the table represents the correlation between two specific variables. This matrix typically computes pairwise correlations primarily using the Pearson correlation coefficient. This coefficient quantifies the linear relationship between variables, ranging from -1 (perfect negative correlation) to 1 (perfect positive correlation), with 0 indicating no correlation. The primary utility of calculating a correlation matrix in feature selection processes for machine learning models lies in its ability to identify columns within the DataFrame that exhibit the highest absolute correlations. This identification is crucial as features with high correlation may convey redundant information. Consequently, eliminating one of such highly correlated features can significantly reduce the dimensionality of the model without substantial loss of information.

In data analysis, particularly when dealing with complex datasets, visualizing the full correlation matrix is often impractical due to its extensive size. To effectively manage this challenge and enhance the clarity of data visualization, we focus on selecting the most informative columns by identifying those with the highest absolute correlation values. This approach allows us to distill the correlation matrix to its most significant relationships, simplifying the visualization and making it more accessible for interpretation while preserving the critical insights from the larger dataset.

The correlation matrix depicted in Figure 4 illustrates the relationships between various features in the XAPI dataset, reflecting educational data dynamics. Notably, Nationality and Place of Birth exhibit a high positive correlation (0.8), indicating a strong alignment between students' nationalities and their birthplaces. Conversely, StageID and GradeID demonstrate a strong negative correlation ($-1.0$), suggesting an inverse relationship possibly due to the dataset's educational stage and grade level encoding. Academic engagement indicators such as raisedhands, VisITedResources and AnnouncementsView positively correlate with each other and with the final class feature, highlighting that increased student

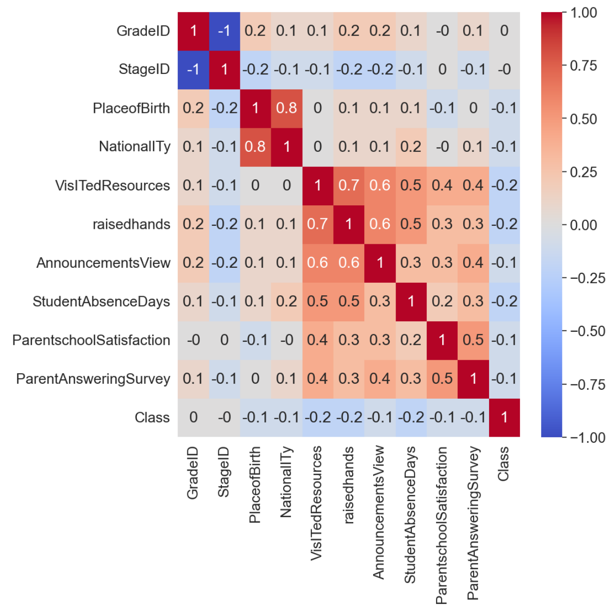



**FIGURE 4.** Correlation heatmap of dataset XAPI.

engagement is associated with better academic performance. Moreover, ParentAnsweringSurvey and ParentschoolSatisfaction share a moderate positive correlation (0.5), revealing that parent satisfaction with the school correlates with their participation in surveys, which also relates positively to the Relation feature, suggesting a link between parent-school relationship quality, satisfaction, and survey participation.

The correlation matrix presented in Figure 5 represents the interrelations among various factors in the SSP dataset, hinting at the complexities within student socio-personal dynamics. Notable observations include the positive correlation between age and failures (0.3), indicating that older students tend to have more academic failures, potentially reflecting challenges like grade repetition. The matrix reveals a strong positive correlation between Medu (mother's education) and Fedu (father's education) at 0.6, underscoring the aligned educational backgrounds within families. Interestingly, the job sectors of mothers (Mjob) and fathers (Fjob) are moderately correlated (0.2), suggesting some level of similarity in parental occupations. Another key

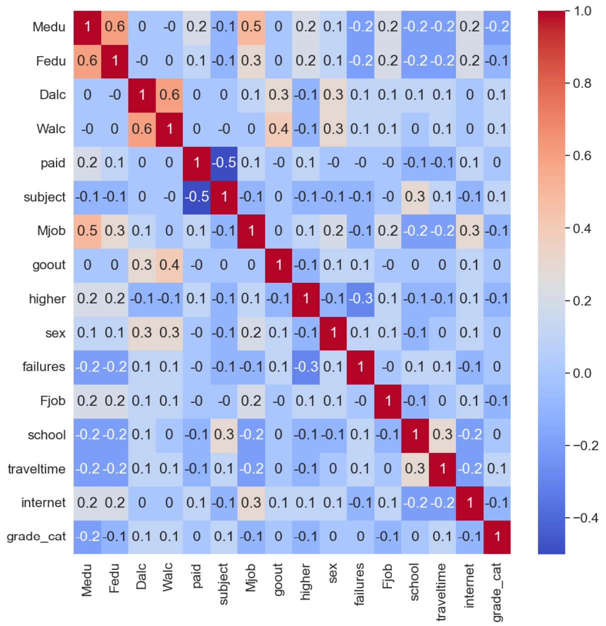



**FIGURE 5.** Correlation heatmap of dataset SSP.

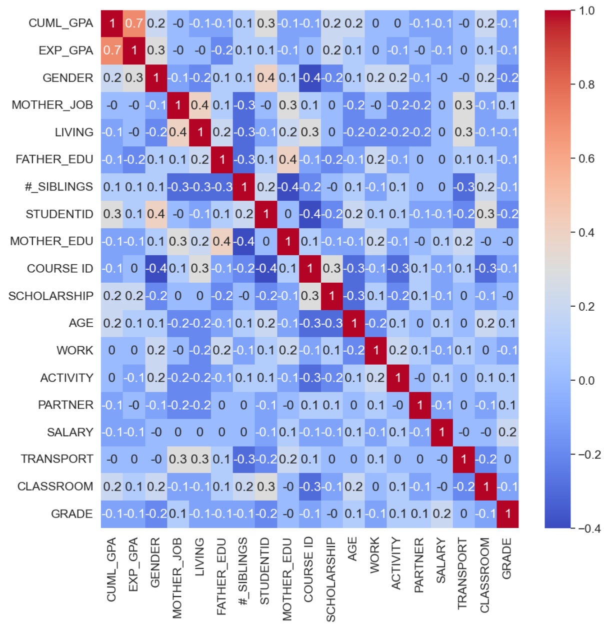



**FIGURE 6.** Correlation heatmap of dataset HESP.

insight is the positive correlation between going out (goout) and weekend alcohol consumption (Walc) at 0.4, which could imply social dynamics influencing drinking habits among students.

A significant negative correlation is observed between paid and subject at −0.5, which might indicate that students taking paid classes do not always correlate with subject, suggesting nuances in the effectiveness or necessity of these classes. The relationship between travel time (traveltime) and school choice (school) is positively marked at 0.3, indicating students attending a specific school might face longer commutes. Additionally, there is correlation between desires for higher education (higher) and internet access (internet) at 0.1, pointing towards the role of digital connectivity in educational aspirations. This matrix, therefore, provides a multifaceted view of the factors affecting students' academic and personal lives, highlighting the influence of family background, personal habits, and educational support on their experiences and outcomes.

The correlation matrix depicted in Figure 6 analyzes various factors related to higher education students' performance (HESP), showcasing how different variables interact within an academic context. Gender shows a good correlation with expected GPA (0.3), suggesting demographic patterns in educational pathways. A negative correlation between scholarship status and age (−0.3) could indicate younger students are more likely to receive scholarships. Work experience positively correlates with gender (0.2) and negatively with scholarship (−0.2), highlighting potential financial independence or necessity among male students.

Activities show a negative correlation with expected GPA (−0.3), suggesting extracurricular engagement might impact academic expectations. Partner status has a slight positive effect on salary (0.1) and living arrangements (0.3), indicating relationship status may influence financial and living conditions. Transport and living conditions share a positive correlation (0.3), reflecting geographical and socio-economic factors affecting student life.

Notably, cumulative GPA (CUML_GPA) positively correlates with expected GPA (EXP_GPA, 0.7), indicating alignment between students' performance expectations and outcomes. However, a strong negative correlation between classroom engagement and grade (−0.3) suggests that active classroom participation does not always correlate with higher grades, a counterintuitive finding that warrants further investigation.

Overall, this matrix reveals complex interdependencies among socio-demographic factors, educational background, personal life, and academic performance in higher education, highlighting the multifaceted nature of student success.

The correlation matrix for the WOC2 dataset, presented in Figure 7, provides insight into the relationships between various academic assessments and the final class performance of students. The matrix highlights that none of the initial quizzes (Quiz01) show a significant correlation with the final class performance, suggesting that these early assessments may not be predictive of final outcomes. However, there is a notable progression in the significance of assignments and exams over time towards class performance.

Assignment01 shows a moderate correlation with Assignment02 (0.6) and Assignment03 (0.6), indicating a consistency in students' performance across assignments. The Midterm Exam holds a stronger correlation (0.4) with the final class, highlighting its predictive value for final

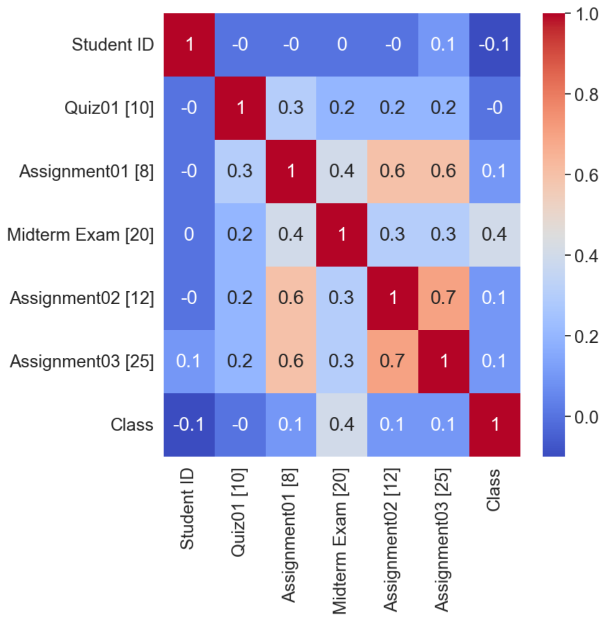



**FIGURE 7. Correlation heatmap of dataset WOC2.**

performance. Interestingly, Assignment03, which has the highest weight (25), also shows a strong correlation with Assignment02 (0.7) and a moderate correlation (0.1) with the final class performance, underlining the importance of consistent performance in higher-weight assignments for overall success.

The matrix also suggests that while quizzes and early assignments may not directly predict the final class performance, they are closely related to each other, hinting at an underlying pattern of student engagement and understanding. This nuanced view provided by the correlation matrix underscores the complexity of academic performance, where later, more heavily weighted assessments have a more pronounced impact on the final class grade, reflecting the cumulative nature of learning and assessment in the educational process.

#### 4) OUTLIER IDENTIFICATION

Outliers, indicative of data points significantly deviating from the overall pattern, were systematically identified and scrutinized. Various techniques, including box plots, z-score analysis, and visualization tools, were employed to detect anomalies that could impact the robustness of machine learning models. The nuanced understanding of outliers obtained during the EDA informed subsequent outlier-handling strategies in the data preprocessing phase.

#### 5) RELATIONSHIPS BETWEEN VARIABLES

An exhaustive analysis of the relationships between variables and the target variable was undertaken. Correlation matrices, bivariate analysis, and statistical tests were employed to discern dependencies and associations, guiding feature selection decisions. This phase was instrumental in unraveling the complex interplay between different attributes and their collective influence on the target variable.

#### 6) DATASET HETEROGENEITY AND FORMATTING CHALLENGES

The datasets under consideration showcased heterogeneity in terms of features, with a diverse range spanning demographic, behavioral, academic, and health-related variables. However, not all datasets were homogeneously formatted for classification tasks. Variations in the number of target classes and class imbalances were identified, necessitating a nuanced approach in the subsequent data preprocessing stage to address these challenges effectively. The insights gleaned from this extensive EDA not only facilitated a profound understanding of the dataset's intricacies but also laid the foundation for informed decisions in subsequent stages of the study. The interplay of statistical analyses, visualizations, and semantic explorations during the EDA phase enriched the dataset's interpretability, contributing significantly to the overall efficacy and accuracy of the subsequent machine-learning endeavors.

### *C. DATA PREPROCESSING*

The data preprocessing phase in this study involved a meticulous series of steps aimed at refining the dataset to enhance its suitability for machine learning applications. Each step was strategically designed to address specific challenges, ensuring the robustness and reliability of subsequent analyses.

#### 1) HANDLING CATEGORICAL DATA

A critical aspect of the data preprocessing process was the treatment of categorical data. To enable the seamless integration of the dataset with machine learning algorithms, categorical variables were encoded using Label Encoders. This transformation facilitated the representation of categorical information in a numerical format, allowing for effective utilization in predictive models.

#### 2) STRATIFIED DATA SPLITTING

To ensure robust model evaluation, the dataset was divided into training and testing sets with an 80-20 split ratio. Stratified sampling was employed during this process to maintain the distribution of target classes in both the training and testing sets. This approach is particularly valuable when dealing with imbalanced datasets, as it preserves the proportionality of class instances, preventing skewed model performance assessments.

#### 3) STANDARDISATION

Standardization was applied to bring all features to a common scale, mitigating the impact of differing measurement units. This step is crucial for algorithms that are sensitive to the scale of input features, ensuring that each feature contributes proportionally to the model's learning process. Standardizing the data fosters improved convergence and stability in machine learning algorithms.

#### 4) MITIGATING CLASS IMBALANCES WITH ADASYN

Class imbalances, where certain classes have significantly fewer instances, pose challenges in classification tasks. To address this, we employ the Adaptive Synthetic Sampling (ADASYN) technique. ADASYN dynamically generates synthetic samples for the minority class based on local density, effectively balancing class distribution and preventing model bias towards the majority class. This approach, as presented by He et al. [66], adapts to the difficulty level of learning for each minority class instance, improving classification by reducing bias and shifting the decision boundary towards challenging examples.

#### 5) TRANSFORMING SSP AND HESP DATASETS WITH THE FIVE-NUMBER METHOD

The SSP and HESP datasets underwent a unique transformation using the five-number method to address specific characteristics. In this method, the class counts within these datasets were divided into three categories: "Low," "Mid," and "High." This categorization aimed to provide a nuanced representation of the class distribution, facilitating a more granular evaluation of model performance across different levels of class representation. The adoption of the five-number method in these datasets was driven by the need for a tailored approach to handling class distributions, acknowledging the nuances inherent in the SSP and HESP datasets. The intricate combination of these preprocessing techniques laid the groundwork for subsequent stages of the study, ensuring that the dataset was optimized in terms of structure, scale, and class distribution for effective machine learning model development and evaluation.

### D. FEATURE SELECTION METHODS

The feature selection phase is integral to optimizing machine learning models by streamlining the input space, enhancing efficiency, and mitigating the risk of overfitting. In this section, we delve into a diverse set of feature selection algorithms, each designed to strategically identify and retain the most informative attributes. Recursive Feature Elimination (RFE) systematically removes less impactful features, refining the model's input set. Forward Selection (FS) incrementally builds the feature subset based on predictive performance, iteratively enhancing model accuracy. Genetic Algorithm (GE) employs evolutionary principles to evolve a subset of features, mimicking the process of natural selection to arrive at an optimal feature configuration. Additionally, we introduce our novel contribution, the Adaptive Feature Selection Algorithm (AFSA), which adapts dynamically to the dataset's characteristics, offering a tailored and efficient approach to feature selection. Through a comparative exploration of these methods, we aim to unravel their efficacy in enhancing model performance and contributing to the overall success of the machine learning endeavors in this study.

#### 1) RECURSIVE FEATURE ELIMINATION (RFE)

Recursive Feature Elimination (RFE) stands as a powerful feature selection algorithm designed to iteratively enhance model performance by systematically eliminating the least informative features. The fundamental principle underlying RFE involves a recursive process that begins with the entire feature set. In each iteration, a model is trained on the current set of features, and the features are then ranked based on a specified criterion, often tied to their contribution to predictive accuracy. Subsequently, the feature with the lowest ranking is systematically removed. This recursive cycle continues until the algorithm converges to the optimal subset of features, determined by predefined criteria or through cross-validation. The algorithm's iterative nature allows it to capture the evolving dynamics of feature importance, ensuring that the most relevant attributes are retained in the final subset. This adaptability makes RFE particularly valuable in scenarios where the initial feature set might be extensive, containing redundant or irrelevant information. The iterative refinement process not only contributes to computational efficiency but also guards against overfitting by focusing on the most informative features. Algorithmically, the RFE process involves several key steps, as outlined in Algorithm 1. The initial model is trained on the complete feature set, and at each iteration, the least informative feature is identified and eliminated. The algorithm converges when the specified criteria for feature subset optimization are met. Through this nuanced process, RFE serves as a sophisticated tool for enhancing the efficiency and predictive power of machine learning models by strategically paring down the feature space to its most informative components.

#### 2) FORWARD SELECTION (FS)

Forward Selection (FS) emerges as a dynamic feature selection algorithm, distinguished by its incremental approach to constructing a feature subset through the iterative addition of the most informative features. Unlike backward elimination strategies, FS initiates with an empty feature set and progressively augments it until a predetermined termination criterion is satisfied. This criterion is typically tied to optimizing model performance, whether through maximizing predictive accuracy or adhering to a specific evaluation metric. The fundamental premise of FS lies in its adaptability and ability to navigate through the feature space, strategically incorporating attributes that contribute most significantly to the model's efficacy. This stepwise refinement process ensures that the algorithm converges to a feature subset optimized for the given task, steering clear of redundant or less impactful attributes. Algorithm 2 outlines the procedural steps of the Forward Selection algorithm. It commences with an empty feature set, and at each iteration, evaluates the impact of adding each remaining feature. The feature that

**Algorithm 1** Recursive Feature Elimination (RFE)

```
Input: Input DataFrame, Target Column, Model, Number of
    Folds for Cross-Validation (cv)
Output: Selected Features
X ← Input DataFrame
y ← Target column of DataFrame
selected_features ← set of all features
n ← number of features in selected_features
optimal_score ← 0
optimal_feature_subset ← Ø
while true do
    worst_feature ← None
    for feature in selected_features do
        candidate_features ← selected_features − feature
        X_train_fs ← X[candidate_features]
        cv_accuracy_fs ← cross_val_score(Model,
        X_train_fs, y, cv, 'accuracy')
        avg_accuracy ← mean(cv_accuracy_fs)
        if avg_accuracy > optimal_score then
            optimal_score ← avg_accuracy
            optimal_feature_subset ← candidate_features
            worst_feature ← feature
        end if
    end for
    if worst_feature is not None then
        selected_features ← selected_features −
        worst_feature
        n ← n - 1
        if n ≤ 1 then
            break
        end if
    else
        break
    end if
end while
```

**Algorithm 2** Forward Selection (FS)

```
Input: Input DataFrame, Target Column, Model, Number of
    Folds for Cross-Validation (cv)
Output: Selected Features
X ← Input DataFrame
y ← Target column of DataFrame
selected_features ← Ø
n ← number of features in X
optimal_score ← 0
optimal_feature_subset ← Ø
while true do
    best_feature ← None
    for feature in X do
        if feature not in selected_features then
            candidate_features ← selected_features + fea-
            ture
            X_train_fs ← X[candidate_features]
            cv_accuracy_fs ← cross_val_score(Model,
            X_train_fs, y, cv, 'accuracy')
            avg_accuracy ← mean(cv_accuracy_fs)
            if avg_accuracy > optimal_score then
                optimal_score ← avg_accuracy
                best_feature ← feature
            end if
        end if
    end for
    if best_feature is not None then
        selected_features ← selected_features +
        best_feature
        n ← n - 1
        if n ≤ 0 then
            break
        end if
    else
        break
    end if
end while
```

maximizes the specified criterion is selected for inclusion, and this process iterates until the termination criterion is met. Forward Selection is particularly advantageous in scenarios where the initial feature set is extensive, and computational efficiency is a priority. By strategically building the feature subset based on the most informative attributes, FS offers a tailored and efficient approach to feature selection, contributing to the overall optimization of machine learning models.

### 3) GENETIC ALGORITHM (GA)

A genetic algorithm for feature selection, presented in Algorithm 3, is a heuristic optimization technique inspired by the process of natural selection and evolution. It is employed to automatically determine the most relevant subset of features from a given dataset for use in machine learning models. The primary objective is to enhance model performance, reduce overfitting, and enhance interpretability by identifying and retaining the most informative features while discarding irrelevant or redundant ones. In this algorithm, a population of potential feature subsets is initialized, typically represented as binary strings, where each bit corresponds to the inclusion or exclusion of a specific feature. The algorithm then evaluates the fitness of these subsets by training and cross-validating a machine-learning model with each feature combination. Fitness is typically measured in terms of the model's performance on a specified metric, such as accuracy or F1-score. The fittest feature subsets, those that yield the best model performance, are selected to form a new generation. Genetic operations like crossover and mutation are applied to the selected individuals to create a new population. Crossover combines two feature subsets to produce offspring with a mixture of their parent's characteristics. Mutation introduces small random changes to the feature subsets to maintain genetic diversity. This iterative process continues for a fixed number of generations, allowing the algorithm to converge towards a feature subset that optimizes model performance. Ultimately, the algorithm outputs the best-selected features, which can then be used to train a final machine-learning model with improved generalization and efficiency.

**Algorithm 3** Genetic Algorithm for Feature Selection

**Input:** Input DataFrame, Target Column, Population Size, Number of Generations, Mutation Rate, Model, Number of Folds for Cross-Validation (cv)
**Output:** Best Feature Subset

```
X ← Input DataFrame
y ← Target column of DataFrame
n ← number of features in X
population ← randomly initialize population with binary feature selection masks
for generation ← 1 to Number of Generations do
    scores ← empty list
    for individual in population do
        selected_features ← features selected based on the individual's mask
        X_train_fs ← X[selected_features]
        cv_accuracy ← cross_val_score(Model, X_train_fs, y, cv, 'accuracy')
        Append cv_accuracy to scores
    end for
    best_indices ← indices of the top-performing individuals
    best_population ← individuals at best_indices
    best_individual ← individual with the highest accuracy among best_population
    new_population ← best_population
    while size of new_population < Population Size do
        parent1, parent2 ← randomly select two individuals from best_population
        crossover_point ← randomly select a crossover point
        child ← combine parent1 and parent2 using crossover at crossover_point
        Apply mutation with probability Mutation Rate to child (flip some bits)
        Append child to new_population
    end while
    population ← new_population
end for
best_selected_features ← features selected based on best_individual's mask
return best_selected_features
```

### 4) ADAPTIVE FEATURE SELECTION ALGORITHM (AFSA)

The Adaptive Feature Selection Algorithm (AFSA), as delineated in Figure 8 and Algorithm 4, represents a sophisticated approach to feature selection, combining multiple ranking methods to ascertain the relevance of each feature. The algorithm employs a diverse array of feature ranking methods, including Information Gain, Chi-Square Test, Mutual Information, Relief, and Gini Importance. Each feature's relevance is quantified through these methods, followed by the normalization of ranks and computation of an average rank for individual features. The core of AFSA lies in its iterative feature selection process, where features are evaluated and selected based on their rank order. The algorithm begins with the top-ranked feature and progressively adds features to the subset, each time assessing the model's cross-validation accuracy. A unique aspect of this process is the incorporation of the *is_improvement_good_enough* function, which introduces a dynamic threshold for determining the significance of accuracy improvement upon the inclusion of new features. This function operates under the premise that as the algorithm proceeds down the list of ranked features, the criterion for improvement becomes less stringent. It begins with a 5% threshold for top-ranked features, decrementing by 0.5% for each subsequent feature. However, the minimum threshold is set at 1%, ensuring that only meaningful improvements contribute to the retention of a feature. This dynamic thresholding serves a dual purpose: it not only seeks to enhance the model's performance by including impactful features but also ensures the compactness of the feature set by prioritizing significant improvements, especially for higher-ranked features. In essence, AFSA is designed to reduce dataset dimensionality by adaptively selecting features that offer the most information. This combination of diverse feature ranking methods with an iterative, threshold-based selection strategy enables AFSA to conduct a comprehensive assessment of feature relevance. Such an approach not only enhances the accuracy of machine learning models but also contributes to their efficiency by focusing on the most influential features, thereby streamlining the predictive process.

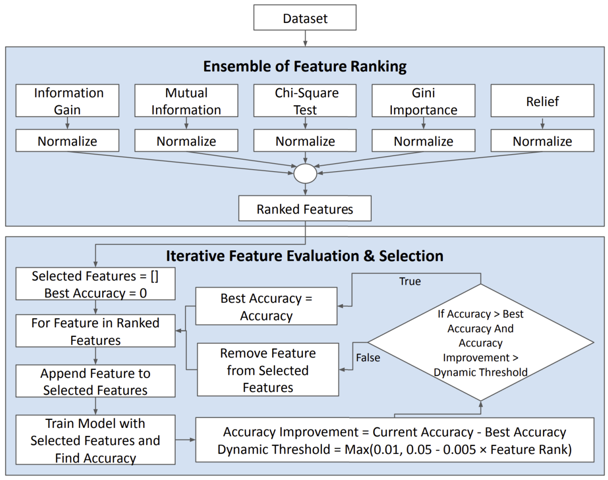



**FIGURE 8.** Flowchart of adaptive feature selection algorithm (AFSA).

**Algorithm 4** Adaptive Feature Selection Algorithm (AFSA)

**Input:** X_train, X_test, y_train, y_test, model
**Output:** Selected Features, Evaluation Metrics

```
Define function is_improvement_good_enough(last_accuracy, current_accuracy, feature_position):
    threshold ← max(0.01, 0.05 - 0.005 × feature_position)
    return current_accuracy - last_accuracy > threshold
Define function get_feature_rank(DataFrame, TargetColumn):
    Calculate feature scores using Information Gain, Chi-Square Test, Mutual Information, Relief, and Gini Importance
    Rank the features based on feature scores
    Normalize each rank and calculate the average rank
    Sort features based on the average rank in descending order
    return sorted list of features
sorted_features ← get_feature_rank(Input DataFrame, Target Column)
selected_features ← empty list
best_accuracy ← 0
for feature in sorted_features do
    selected_features.append(feature)
    X_train_fs ← X_train[selected_features]
    X_test_fs ← X_test[selected_features]
    cv_accuracy ← mean(cross_val_score(model, X_train_fs, y_train, 'accuracy'))
    if cv_accuracy > best_accuracy and is_improvement_good_enough(best_accuracy, cv_accuracy, idx) then
        best_accuracy ← cv_accuracy
    else
        selected_features.remove(feature)
    end if
end for
Train the model with selected features and evaluate on the test set
Calculate evaluation metrics including accuracy, F1 score, precision, and recall for both cross-validation and test sets
Record execution time and list of selected features
return selected_features, evaluation metrics
```

#### 5) DELINEATING AFSA FROM FORWARD SELECTION: A NOVELTY PERSPECTIVE

To address the distinctiveness of the Adaptive Feature Selection Algorithm (AFSA) from the traditional Forward Selection (FS) approach, it is imperative to underscore the foundational and operational differences that underscore AFSA's novelty and its contribution to the existing corpus of knowledge on feature selection methodologies. While both strategies aim at enhancing model performance through judicious feature selection, the mechanisms they employ to achieve this objective exhibit significant divergences.

##### *a: METHODOLOGICAL DISTINCTION*

The crux of Forward Selection lies in its incremental nature, beginning with an empty set and progressively adding features that maximize a performance criterion until no further improvements can be made. This process, although systematic, does not incorporate multiple feature ranking metrics or a dynamic thresholding mechanism for feature inclusion. Conversely, AFSA integrates a preliminary comprehensive feature ranking phase, employing multiple evaluation metrics such as Information Gain, Chi-Square Test, Mutual Information, Relief, and Gini Importance. This multifaceted ranking not only provides a robust basis for feature evaluation but also enhances the algorithm's adaptability to varied data characteristics.

##### *b: ADAPTIVE SELECTION PROCESS*

Beyond the initial ranking, AFSA distinguishes itself through an adaptive feature inclusion strategy. Unlike FS, which adheres to a static metric for feature addition, AFSA utilizes a dynamic thresholding function (*is_improvement_good_enough*) to determine the significance of including a new feature based on its position in the ranked list. This adaptability allows AFSA to maintain a balance between model complexity and performance, ensuring the inclusion of features that offer substantial improvement and discarding those with marginal benefits.

##### *c: COMPUTATIONAL EFFICIENCY*

The operational efficiency of AFSA further sets it apart from FS. By leveraging the ranked features list and dynamically adjusting the inclusion criteria, AFSA minimizes unnecessary model evaluations, reducing computational time. This efficiency is particularly crucial in big data contexts, where the volume of features can significantly impact the feasibility of feature selection processes.

##### *d: CONTRIBUTION TO KNOWLEDGE*

The amalgamation of these features–multi-criteria ranking, adaptive thresholding, and enhanced computational efficiency–positions AFSA as a significant advancement over traditional FS. It not only broadens the methodological toolkit available for feature selection but also introduces a nuanced approach that addresses some of the limitations inherent in existing strategies. Through these innovations, AFSA contributes to the body of knowledge by offering a versatile, efficient, and effective solution for feature selection, applicable across a diverse array of machine learning tasks and data contexts.

In summary, the distinction between AFSA and FS is not merely a matter of implementation details but a fundamental difference in approach, efficiency, and adaptability. By providing a detailed exposition of these differences, we aim to clarify the unique contributions of AFSA to the field of feature selection in machine learning.

#### 6) RATIONALE BEHIND AFSA'S HYBRID APPROACH

The inception of AFSA was driven by a pivotal goal: to amalgamate the efficiency and expediency inherent in filter-based feature selection approaches with the precision and performance optimization characteristic of more iterative methodologies, such as Forward Selection and Recursive Feature Elimination. This hybrid strategy stems from a critical observation: while filter-based methods excel in reducing training time due to their simplicity and independence from learning algorithms, they often do not account for feature dependencies or interaction effects, potentially compromising predictive accuracy. Conversely, methods like Forward Selection, though highly effective in optimizing model performance through exhaustive feature evaluation, can be prohibitively time-consuming and computationally intensive, especially for datasets with a high-dimensional feature space.

The AFSA framework was thus designed to bridge this gap, leveraging the speed of filter-based approaches through an initial feature ranking phase that employs a diverse set of metrics, including Information Gain, Chi-Square Test, Mutual Information, Relief, and Gini Importance. This phase serves to quickly reduce the dimensionality of the dataset, focusing subsequent analysis on features most likely to impact model performance. The iterative selection process that follows, incorporating a dynamic threshold for accuracy improvement, mirrors the thoroughness of more computationally demanding methods. By adjusting the improvement threshold based on feature ranking, AFSA ensures that early additions must meet a higher standard of performance enhancement, reflecting their presumed greater impact, while allowing for more flexibility as the selection process progresses. This nuanced approach ensures that AFSA remains both swift and effective, marrying the rapid training times of filter-based methods with the accuracy and model performance enhancements typical of iterative selection techniques.

Empirical evidence from our experiments substantiates the efficacy of this approach. Training time comparisons between AFSA and traditional methodologies underscore the significant reductions in computational overhead achieved by AFSA, without sacrificing model accuracy. These findings are particularly compelling, demonstrating that AFSA provides a viable pathway to achieving high-performance machine learning models in a fraction of the time required by conventional methods. Through this innovative amalgamation of speed and precision, AFSA represents a significant advancement in the field of feature selection, offering a scalable and efficient solution for high-dimensional data analysis.

#### 7) THEORETICAL COMPUTATIONAL EFFICIENCY ANALYSIS

To thoroughly evaluate the computational efficiency of our proposed Adaptive Feature Selection Algorithm (AFSA) relative to established feature selection methods such as Recursive Feature Elimination (RFE), Forward Selection (FS), and Genetic Algorithms (GA), we embark on a theoretical analysis of their respective time complexities. This analysis not only clarifies the inherent computational demands of each algorithm but also positions AFSA within the context of big data applications, highlighting its potential efficiency advantages.

##### *a: RECURSIVE FEATURE ELIMINATION (RFE)*

The time complexity of RFE is influenced by the number of features $n$ and the iterative process of feature elimination. Assuming a constant time complexity $O(1)$ for each model training and evaluation, RFE exhibits a worst-case time complexity of $O(n^2)$. This quadratic complexity arises from the cumulative training and evaluation steps across $n$ iterations, indicative of a significant computational burden for large feature sets.

##### *b: FORWARD SELECTION (FS)*

FS operates in a manner akin to RFE but in reverse, adding features incrementally. Under the same assumption of constant evaluation time, FS's time complexity similarly approximates to $O(n^2)$. This reflects the iterative addition and evaluation of features, rendering FS computationally intensive, especially as the number of features grows.

##### *c: GENETIC ALGORITHM (GA)*

The computational complexity of GA hinges on the population size $p$, the number of generations $g$, and the fitness evaluation process, predominantly the model training for each feature subset. Assuming a linear complexity for fitness evaluation, the overall complexity of GA can be expressed as $O(p \cdot g \cdot n)$, where $n$ denotes the number of features. This suggests that GA's computational demand scales with the size of the feature set and the parameters governing the evolutionary process.

##### *d: ADAPTIVE FEATURE SELECTION ALGORITHM (AFSA)*

The Adaptive Feature Selection Algorithm (AFSA) employs a distinctive hybrid approach, initially leveraging multiple ranking methods for a comprehensive evaluation of features, followed by an adaptive, iterative selection process. The algorithm's complexity primarily stems from two components: the feature ranking and the selection phases. Given the absence of feature reduction or filtering in the initial ranking, and considering the use of multiple ranking methods, the complexity of this phase can be approximated as $O(n \log n)$, assuming the ranking algorithms employed operate in logarithmic time with respect to the number of features $n$. The subsequent selection phase, characterized by a single iteration over the ranked features without nested loops, implies a linear complexity, $O(n)$. Therefore, AFSA's overall complexity is more accurately described by $O(n \log n + n)$, which simplifies to $O(n \log n)$. This adjustment underscores AFSA's efficiency, particularly highlighting its capability to process large feature sets with a linearly scalable selection phase, thereby minimizing the computational overhead typically associated with feature selection in high-dimensional datasets.

Through this theoretical exploration, it becomes evident that while RFE and FS face scalability challenges due to their quadratic complexities, and GA's efficiency is contingent upon evolutionary parameters, AFSA's design inherently aims at reducing computational overhead. This analysis highlights AFSA's potential to deliver a balanced and efficient approach to feature selection in the realm of big data.

#### 8) COMPARATIVE ANALYSIS OF AFSA

Unlike traditional methods, AFSA integrates a multifaceted ranking mechanism and an adaptive threshold-based selection strategy, distinguishing itself through the amalgamation of robustness and efficiency in feature selection.

AFSA's methodology significantly deviates from singular metric approaches like RFE and FS by employing a diverse array of feature ranking methods (Information Gain, Chi-Square Test, Mutual Information, Relief, and Gini Importance). This multiplicity ensures a comprehensive evaluation of feature relevance, mitigating the risk of bias toward specific data characteristics. Consequently, AFSA facilitates a more informed and versatile feature selection process, enhancing model performance across varied datasets.

Moreover, AFSA introduces an adaptive feature selection strategy that dynamically adjusts the inclusion threshold based on the incremental value each feature adds to model accuracy. This approach starkly contrasts with the static nature of FS and the exhaustive retraining in RFE, where features are added or eliminated without adjusting for the diminishing returns on model performance. By implementing a decreasing threshold for accuracy improvement, AFSA optimizes the selection process, effectively balancing between model accuracy and feature set compactness without necessitating extensive retraining at each iteration.

The computational complexity of AFSA, while initially seeming higher due to the utilization of multiple ranking methods, is offset by its strategic iterative selection process. The early termination criteria based on dynamic thresholding significantly reduce the number of model retraining instances, especially in big data contexts. This adaptive mechanism ensures that only features contributing meaningful improvements are considered, thereby minimizing computational overhead. In contrast, traditional methods like RFE may involve numerous retraining steps as they do not adjust selection criteria based on the feature's order or potential impact, leading to higher computational demands.

In summary, AFSA represents a significant advancement in feature selection methodologies by providing a balanced approach that not only offers competitive accuracy and efficient feature identification but also addresses the critical aspect of the computational burden. This enhanced comparative analysis underscores AFSA's theoretical and practical superiority, particularly in handling large-scale datasets, thereby affirming its contribution to the field of machine learning.

### E. CROSS VALIDATION

Cross-validation stands as an indispensable technique in the assessment of machine learning models, playing a pivotal role in gauging their performance robustness and generalisability. This methodology involves the systematic partitioning of the dataset into training and validation sets, with a primary objective of ensuring that every data point is utilized for validation at some point during the evaluation process. This not only enhances the reliability of performance metrics but also guards against potential biases that may arise from a static train-test split. A widely adopted form of cross-validation is the k-fold cross-validation, a process that entails dividing the dataset into k subsets or folds. In each iteration of the cross-validation loop, the model is trained on k-1 folds and validated on the remaining one. This cyclic process is repeated k times, with each fold serving as the validation set exactly once. The results are then averaged across all iterations, providing a comprehensive and unbiased estimate of the model's generalization performance. In the context of this study, where the prediction of student performance serves as a focal point, the application of 5-fold cross-validation was integral. The choice of a 5-fold setup was deliberate, balancing the need for rigorous assessment with computational efficiency. By systematically rotating through different subsets of the dataset during the validation phase, the cross-validation strategy employed in this study contributes to the robustness of model evaluations. Additionally, the use of cross-validation acts as a safeguard against overfitting, ensuring that the models generalize well to unseen data and align with the overarching goals of accuracy and reliability in the prediction of student outcomes.

### F. CLASSIFIERS

In this study, we adopted a selection of classifiers based on the comprehensive survey conducted by Xiao et al. [67] in the field of educational data mining (EDM). Xiao et al. provided a valuable list of widely used models in EDM research, laying the foundation for our model selection. However, the specific criteria for our selection were defined with our specific scenario in mind. Our primary considerations were the prevalence of models in EDM, their lightweight nature for iterative feature selection algorithms, and the exclusion of ensemble models for a fair comparison. The selected classifiers encompass a diverse set of models commonly employed in EDM research. Decision Trees (DT), K-Nearest Neighbours (KNN), Logistic Regression (LR), Naive Bayes (NB), and Support Vector Machines (SVM) were chosen for their widespread usage and compatibility with our defined criteria. DT and NB offer interpretability and swift training and prediction speeds. KNN, known for its simplicity and noise insensitivity, was chosen with consideration for its suitability in smaller datasets. LR, a classical regression algorithm, provides fast training and ease of understanding. SVM, effective in classification, rounds out our selected models. The model selection process, informed by the survey findings of Xiao et al. [67], focused on models widely used in EDM research. The chosen classifiers meet our specific criteria of prevalence, lightweight nature, and exclusion of ensemble models. This strategic selection ensures that our comparative analysis provides meaningful insights into the predictive performance of individual classifiers in the context of student performance prediction using adaptive feature selection algorithms.

### G. EVALUATION

The assessment of the machine learning models employed in this study was conducted using a diverse set of performance metrics, including accuracy, F1-score, precision, recall, and the feature reduction factor (FRF), as detailed in Equation 1. These metrics collectively offered a comprehensive and nuanced evaluation of the model's efficacy in predicting student performance across various scenarios. The adoption of a 5-fold cross-validation approach ensured a robust and reliable assessment of model performance, mitigating the impact of data partitioning on the evaluation outcomes. Accuracy, a fundamental metric, provided an overarching measure of the models' correctness in predictions. The F1-score, precision, and recall offered insights into the models' ability to balance true positives, false positives, and false negatives, providing a more nuanced understanding of performance. The feature reduction factor (FRF) played a distinctive role in quantifying the impact of feature selection on model performance. As defined in Equation 1, the FRF represented the ratio of the total feature count to the selected feature count. This metric served as a valuable indicator of the efficiency of feature selection algorithms, offering insights into the degree of dimensionality reduction achieved.

$$\text{FRF} = \frac{\text{Total feature count}}{\text{Selected feature count}} \tag{1}$$

The adoption of such a comprehensive set of metrics, coupled with the meticulous 5-fold cross-validation approach, contributed to a thorough evaluation of the machine learning models. The outcomes of this evaluation not only validated the models' predictive capabilities but also provided actionable insights for enhancing educational outcomes in the realm of educational data mining.

## V. RESULTS DISCUSSION

In this section, we present the outcomes of our experiments and engage in a comprehensive discussion of the results. Our focus is on evaluating the performance of the proposed Adaptive Feature Selection Algorithm (AFSA) in comparison to other state-of-the-art feature selection algorithms. We conducted experiments across five diverse machine-learning models and four distinct student performance prediction datasets.

### A. PERFORMANCE METRICS AND COMPARISON

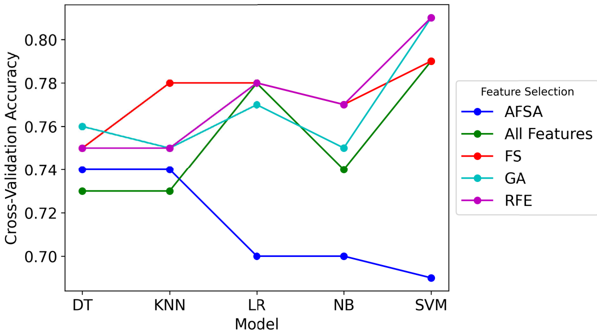
We assessed the effectiveness of various feature selection algorithms by employing cross-validation accuracy, F1 score, precision, recall, feature reduction factor, and training time as key metrics. The comparison included AFSA alongside other widely used feature selection methods. Figure 9 illustrates the cross-validation accuracy for each permutation, organized by feature selection algorithms, across different machine learning models and datasets. Notably, the forward selection (FS) algorithm consistently demonstrated a leading accuracy, with AFSA closely following suit. Other feature selection methods exhibited marginal differences in performance. Concerning the feature reduction factor, Figure 10 demonstrates AFSA's superiority, yielding an average reduction factor of 5.71. In contrast, the best-performing alternative, FS, achieved only 1.8. This underscores AFSA's effectiveness in streamlining datasets without compromising predictive accuracy. The training time analysis, presented in Figure 11, positions AFSA as the top-performing feature selection algorithm, trailing only behind the scenario where no feature selection is applied. AFSA exhibited an average training time of 1.24 seconds, outperforming other algorithms such as Recursive Feature Elimination (RFE), which required 5.81 seconds on average.

### B. DATASET-SPECIFIC PERFORMANCE

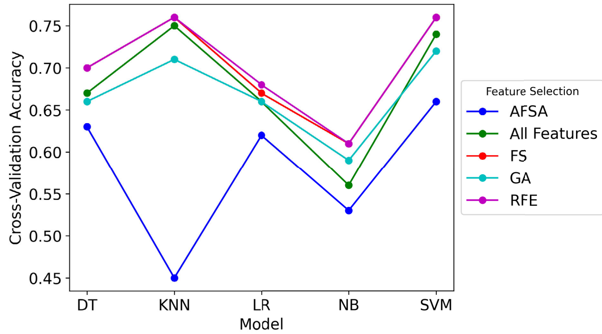
To delve deeper into the performance variations, we present specific evaluations for each dataset. Table 4 showcases the performance metrics for the XAPI dataset across five machine-learning models. Notably, the Genetic Algorithm (GA) excelled for Decision Trees, while Forward Selection (FS) proved optimal for k-Nearest Neighbours. Logistic Regression demonstrated consistent performance across feature selection methods. Support Vector Machines achieved peak accuracy with GA but demonstrated nuanced dependencies on feature selection techniques. Similar analyses were conducted for datasets WOC2, SSP, and HESP, presented in Tables 5, 6, and 7, respectively. These tables provide a comprehensive overview of each model's performance, emphasizing the influence of feature selection techniques on accuracy, precision, recall, and F1 Score, alongside varying training times.

### C. COMPARATIVE ACCURACY ANALYSIS OF FEATURE SELECTION ALGORITHMS

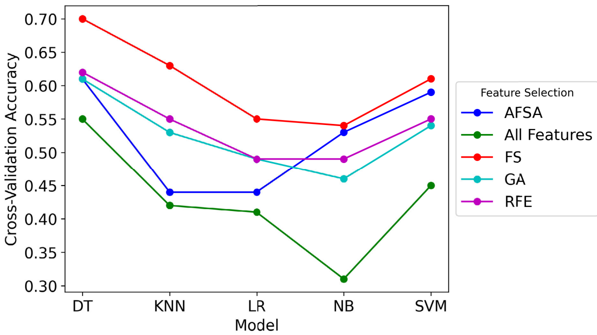
In this part, we delve into a comprehensive comparison of the Adaptive Feature Selection Algorithm (AFSA) against various established feature selection methodologies across multiple models and datasets. Our objective is to elucidate the extent to which accuracy variations manifest, alongside examining the statistical significance of these differences.



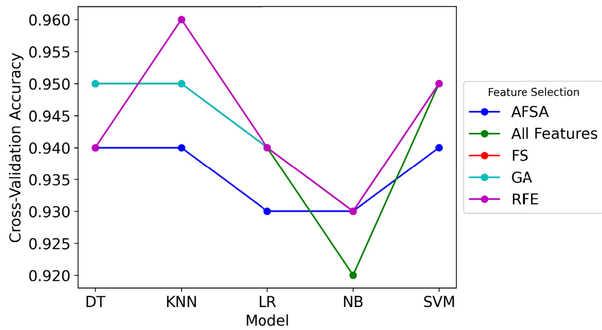

(a) Feature Selection with ML models on XAPI Dataset.




(b) Feature Selection with ML models on SSP Datasets

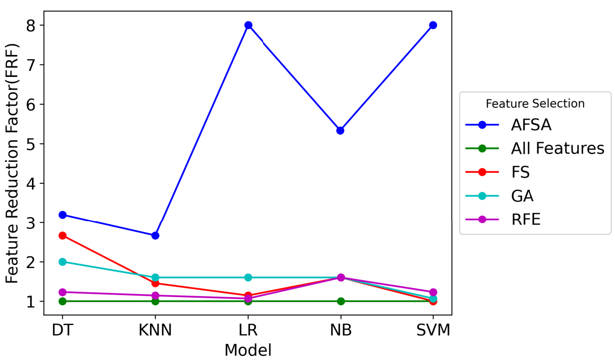



(c) Feature Selection with ML models on HESP Dataset

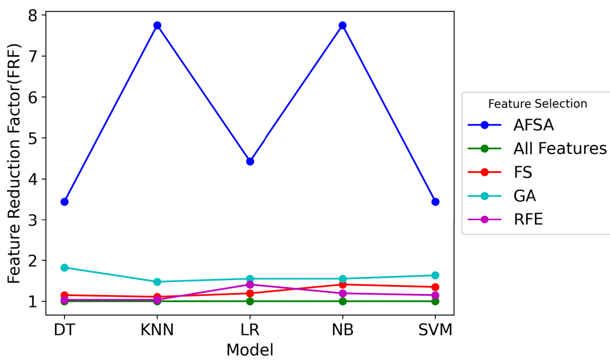



(d) Feature Selection with ML models on WOC2 dataset

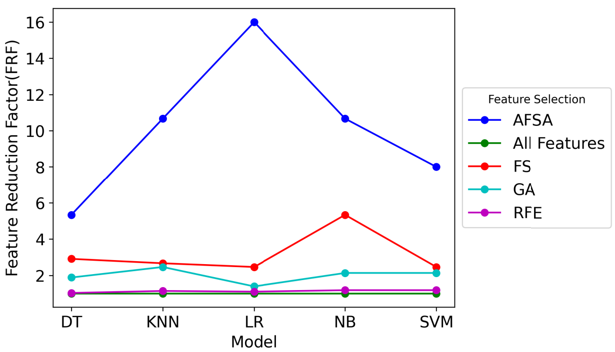
**FIGURE 9.** **Cross-validation accuracy grouped by feature selection with five ML models on four datasets.**



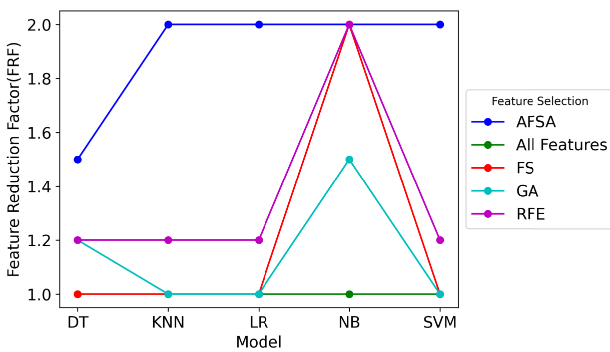

(a) Feature Selection Algorithms with Five models on XAPI Dataset.




(b) Feature Selection Algorithms with Five models on SSP Dataset.

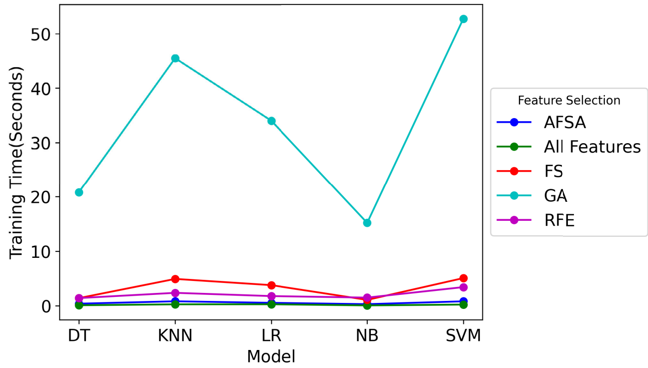



(c) Feature Selection Algorithms with Five models on HESP Dataset

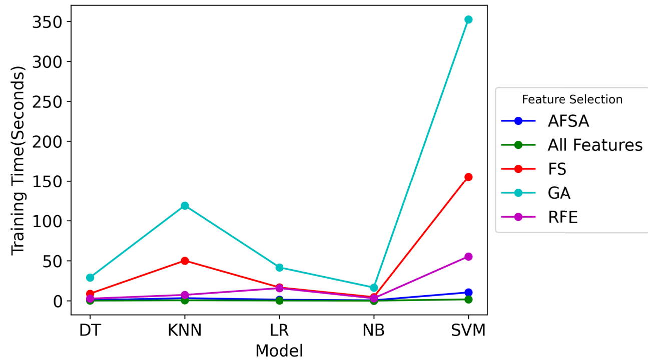



(d) Feature Selection Algorithms with Five models on WOC2 Dataset

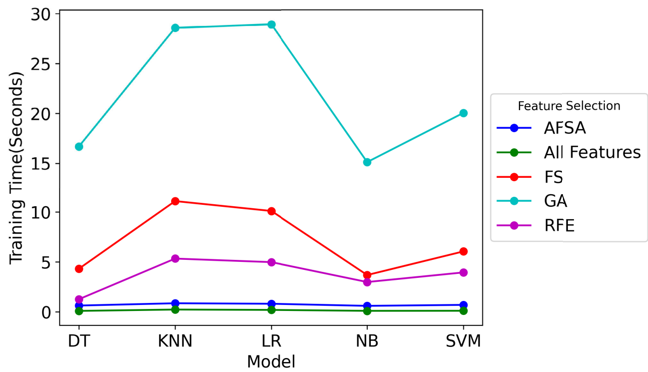
**FIGURE 10.** **Feature reduction factor grouped by feature selection algorithms with five ML models and four datasets.**



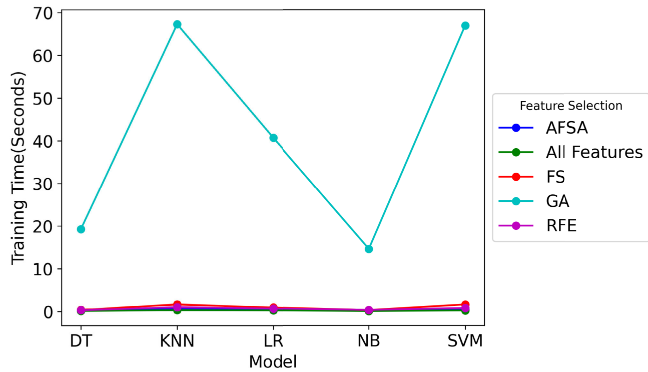

(a) Training time grouped by feature selection algorithms for 5 different machine learning models and dataset XAPI




(b) Training time grouped by feature selection algorithms for 5 different machine learning models and dataset SSP




(c) Training time grouped by feature selection algorithms for 5 different machine learning models and dataset HESP




(d) Training time grouped by feature selection algorithms for 5 different machine learning models and dataset WOC2

**FIGURE 11. Training time grouped by feature selection algorithms for 5 different machine learning models and 4 datasets.**

### 1) VARIATION IN ACCURACY

The analysis reveals notable findings regarding the performance of AFSA in comparison to its counterparts:

- **Maximum Difference in Accuracy:** The apex of AFSA's performance superiority is marked by a maximum accuracy difference of +0.22, indicating scenarios where AFSA outperforms other methods by a margin of 22%.
- **Minimum Difference in Accuracy:** Conversely, the nadir of AFSA's performance is observed at a -0.31 difference, underscoring instances where AFSA lags behind alternative methods by 31% in terms of accuracy.
- **Standard Deviation:** The standard deviation of the accuracy differences between AFSA and the comparative algorithms stands at approximately 0.082, reflecting the variability and consistency of AFSA's performance across diverse scenarios.

### 2) STATISTICAL SIGNIFICANCE ANALYSIS

A detailed statistical significance analysis, employing t-tests to compare the accuracy discrepancies between AFSA and each of the benchmarked feature selection algorithms, yielded the following p-values:

- **All Features (no selection):** A p-value of 0.83 suggests no statistically significant difference in performance when juxtaposing AFSA with the baseline approach of utilizing all features.
- **Feature Selection (FS):** With a p-value of 0.17, the accuracy differential between AFSA and FS does not reach statistical significance, albeit being relatively closer to the threshold of significance.
- **Genetic Algorithm (GA):** The p-value of 0.46 indicates the absence of a statistically significant accuracy difference between AFSA and GA.
- **Recursive Feature Elimination (RFE):** A p-value of 0.32 similarly denotes a lack of statistically significant difference in accuracy between AFSA and RFE.

The comprehensive comparative analysis underscores that while variations in accuracy between AFSA and other feature selection methodologies are observed, these variations generally do not translate to statistically significant differences. Such discrepancies could be attributed more to the specific characteristics inherent to the datasets or models rather than to an intrinsic superiority of one feature selection method over another. The absence of statistically significant differences, especially when compared to the conventional approach of

**TABLE 4.** Evaluation metrics on XAPI dataset.

| Model | Feature Selection | Accuracy | F1 Score | Precision | Recall | FRF | Training Time(s) |
|---|---|---|---|---|---|---|---|
| DT | AFSA | 0.74 | 0.75 | 0.75 | 0.75 | 3.2 | 0.4 |
| DT | All Features | 0.73 | 0.73 | 0.74 | 0.74 | 1 | 0.11 |
| DT | FS | 0.75 | 0.75 | 0.76 | 0.75 | 2.67 | 1.41 |
| DT | GA | 0.76 | 0.76 | 0.76 | 0.76 | 2 | 20.83 |
| DT | RFE | 0.75 | 0.75 | 0.76 | 0.75 | 1.23 | 1.43 |
| KNN | AFSA | 0.74 | 0.75 | 0.75 | 0.75 | 2.67 | 0.84 |
| KNN | All Features | 0.73 | 0.72 | 0.73 | 0.74 | 1 | 0.28 |
| KNN | FS | 0.78 | 0.77 | 0.78 | 0.78 | 1.45 | 4.93 |
| KNN | GA | 0.75 | 0.74 | 0.75 | 0.75 | 1.6 | 45.51 |
| KNN | RFE | 0.75 | 0.74 | 0.74 | 0.75 | 1.14 | 2.38 |
| LR | AFSA | 0.7 | 0.69 | 0.71 | 0.7 | 8 | 0.56 |
| LR | All Features | 0.78 | 0.78 | 0.78 | 0.78 | 1 | 0.29 |
| LR | FS | 0.78 | 0.78 | 0.78 | 0.78 | 1.14 | 3.79 |
| LR | GA | 0.77 | 0.77 | 0.77 | 0.77 | 1.6 | 34.02 |
| LR | RFE | 0.78 | 0.78 | 0.78 | 0.78 | 1.07 | 1.79 |
| NB | AFSA | 0.7 | 0.69 | 0.71 | 0.71 | 5.33 | 0.31 |
| NB | All Features | 0.74 | 0.74 | 0.74 | 0.74 | 1 | 0.08 |
| NB | FS | 0.77 | 0.76 | 0.77 | 0.77 | 1.6 | 1.09 |
| NB | GA | 0.75 | 0.75 | 0.75 | 0.75 | 1.6 | 15.19 |
| NB | RFE | 0.77 | 0.77 | 0.77 | 0.78 | 1.6 | 1.51 |
| SVM | AFSA | 0.69 | 0.69 | 0.72 | 0.71 | 8 | 0.83 |
| SVM | All Features | 0.79 | 0.79 | 0.8 | 0.8 | 1 | 0.24 |
| SVM | FS | 0.79 | 0.79 | 0.8 | 0.8 | 1 | 5.08 |
| SVM | GA | 0.81 | 0.81 | 0.81 | 0.81 | 1.07 | 52.72 |
| SVM | RFE | 0.81 | 0.81 | 0.81 | 0.81 | 1.23 | 3.43 |

**TABLE 5.** Evaluation metrics on WOC2 dataset.

| Model | Feature Selection | Accuracy | F1 Score | Precision | Recall | FRF | Training Time(s) |
|---|---|---|---|---|---|---|---|
| DT | AFSA | 0.94 | 0.94 | 0.94 | 0.94 | 1.5 | 0.2 |
| DT | All Features | 0.95 | 0.95 | 0.95 | 0.95 | 1 | 0.11 |
| DT | FS | 0.95 | 0.95 | 0.95 | 0.95 | 1 | 0.38 |
| DT | GA | 0.95 | 0.95 | 0.95 | 0.95 | 1.2 | 19.27 |
| DT | RFE | 0.94 | 0.94 | 0.94 | 0.94 | 1.2 | 0.32 |
| KNN | AFSA | 0.94 | 0.94 | 0.94 | 0.94 | 2 | 0.65 |
| KNN | All Features | 0.95 | 0.95 | 0.96 | 0.95 | 1 | 0.32 |
| KNN | FS | 0.95 | 0.95 | 0.96 | 0.95 | 1 | 1.65 |
| KNN | GA | 0.95 | 0.95 | 0.96 | 0.95 | 1 | 67.31 |
| KNN | RFE | 0.96 | 0.96 | 0.96 | 0.96 | 1.2 | 1.01 |
| LR | AFSA | 0.93 | 0.94 | 0.94 | 0.94 | 2 | 0.4 |
| LR | All Features | 0.94 | 0.94 | 0.94 | 0.94 | 1 | 0.25 |
| LR | FS | 0.94 | 0.94 | 0.94 | 0.94 | 1 | 0.91 |
| LR | GA | 0.94 | 0.94 | 0.94 | 0.94 | 1 | 40.74 |
| LR | RFE | 0.94 | 0.94 | 0.94 | 0.94 | 1.2 | 0.69 |
| NB | AFSA | 0.93 | 0.93 | 0.93 | 0.93 | 2 | 0.16 |
| NB | All Features | 0.92 | 0.92 | 0.92 | 0.92 | 1 | 0.09 |
| NB | FS | 0.93 | 0.93 | 0.93 | 0.93 | 2 | 0.32 |
| NB | GA | 0.93 | 0.93 | 0.93 | 0.93 | 1.5 | 14.64 |
| NB | RFE | 0.93 | 0.93 | 0.93 | 0.93 | 2 | 0.35 |
| SVM | AFSA | 0.94 | 0.94 | 0.94 | 0.94 | 2 | 0.49 |
| SVM | All Features | 0.95 | 0.95 | 0.95 | 0.95 | 1 | 0.24 |
| SVM | FS | 0.95 | 0.95 | 0.95 | 0.95 | 1 | 1.62 |
| SVM | GA | 0.95 | 0.95 | 0.95 | 0.95 | 1 | 66.98 |
| SVM | RFE | 0.95 | 0.95 | 0.95 | 0.95 | 1.2 | 0.8 |

**TABLE 6.** Evaluation metrics on SSP dataset.

| Model | Feature Selection | Accuracy | F1 Score | Precision | Recall | FRF | Training Time(s) |
|---|---|---|---|---|---|---|---|
| DT | AFSA | 0.63 | 0.63 | 0.64 | 0.64 | 3.44 | 0.74 |
| DT | All Features | 0.67 | 0.67 | 0.68 | 0.67 | 1 | 0.19 |
| DT | FS | 0.7 | 0.7 | 0.7 | 0.7 | 1.15 | 8.9 |
| DT | GA | 0.66 | 0.66 | 0.67 | 0.66 | 1.82 | 29.08 |
| DT | RFE | 0.7 | 0.7 | 0.7 | 0.7 | 1.03 | 2.64 |
| KNN | AFSA | 0.45 | 0.46 | 0.59 | 0.5 | 7.75 | 3.26 |
| KNN | All Features | 0.75 | 0.73 | 0.77 | 0.75 | 1 | 0.55 |
| KNN | FS | 0.76 | 0.75 | 0.78 | 0.76 | 1.11 | 50.09 |
| KNN | GA | 0.71 | 0.69 | 0.73 | 0.71 | 1.48 | 118.96 |
| KNN | RFE | 0.76 | 0.75 | 0.78 | 0.76 | 1.03 | 7.31 |
| LR | AFSA | 0.62 | 0.61 | 0.64 | 0.62 | 4.43 | 1.37 |
| LR | All Features | 0.66 | 0.65 | 0.68 | 0.66 | 1 | 0.28 |
| LR | FS | 0.67 | 0.67 | 0.69 | 0.67 | 1.19 | 16.62 |
| LR | GA | 0.66 | 0.66 | 0.68 | 0.66 | 1.55 | 41.72 |
| LR | RFE | 0.68 | 0.68 | 0.7 | 0.68 | 1.41 | 15.75 |
| NB | AFSA | 0.53 | 0.5 | 0.61 | 0.54 | 7.75 | 0.57 |
| NB | All Features | 0.56 | 0.54 | 0.68 | 0.57 | 1 | 0.09 |
| NB | FS | 0.61 | 0.6 | 0.63 | 0.61 | 1.41 | 4.56 |
| NB | GA | 0.59 | 0.58 | 0.64 | 0.59 | 1.55 | 16.52 |
| NB | RFE | 0.61 | 0.6 | 0.65 | 0.61 | 1.19 | 3.09 |
| SVM | AFSA | 0.66 | 0.65 | 0.67 | 0.66 | 3.44 | 10.42 |
| SVM | All Features | 0.74 | 0.74 | 0.77 | 0.74 | 1 | 1.7 |
| SVM | FS | 0.76 | 0.76 | 0.78 | 0.76 | 1.35 | 155.29 |
| SVM | GA | 0.72 | 0.72 | 0.74 | 0.72 | 1.63 | 352.4 |
| SVM | RFE | 0.76 | 0.76 | 0.78 | 0.76 | 1.15 | 55.27 |

**TABLE 7.** Evaluation metrics on HESP dataset.

| Model | Feature Selection | Accuracy | F1 Score | Precision | Recall | FRF | Training Time(s) |
|---|---|---|---|---|---|---|---|
| DT | AFSA | 0.61 | 0.64 | 0.65 | 0.64 | 5.33 | 0.63 |
| DT | All Features | 0.55 | 0.55 | 0.57 | 0.54 | 1 | 0.09 |
| DT | FS | 0.7 | 0.7 | 0.72 | 0.7 | 2.91 | 4.31 |
| DT | GA | 0.61 | 0.61 | 0.62 | 0.62 | 1.88 | 16.67 |
| DT | RFE | 0.62 | 0.62 | 0.63 | 0.62 | 1.03 | 1.26 |
| KNN | AFSA | 0.44 | 0.53 | 0.54 | 0.54 | 10.67 | 0.86 |
| KNN | All Features | 0.42 | 0.38 | 0.44 | 0.43 | 1 | 0.23 |
| KNN | FS | 0.63 | 0.63 | 0.67 | 0.64 | 2.67 | 11.1 |
| KNN | GA | 0.53 | 0.53 | 0.55 | 0.52 | 2.46 | 28.58 |
| KNN | RFE | 0.55 | 0.52 | 0.6 | 0.55 | 1.14 | 5.33 |
| LR | AFSA | 0.44 | 0.52 | 0.53 | 0.52 | 16 | 0.81 |
| LR | All Features | 0.41 | 0.4 | 0.42 | 0.41 | 1 | 0.19 |
| LR | FS | 0.55 | 0.53 | 0.54 | 0.55 | 2.46 | 10.11 |
| LR | GA | 0.49 | 0.49 | 0.5 | 0.49 | 1.39 | 28.93 |
| LR | RFE | 0.49 | 0.47 | 0.48 | 0.48 | 1.1 | 4.98 |
| NB | AFSA | 0.53 | 0.54 | 0.54 | 0.57 | 10.67 | 0.6 |
| NB | All Features | 0.31 | 0.28 | 0.34 | 0.31 | 1 | 0.1 |
| NB | FS | 0.54 | 0.55 | 0.59 | 0.54 | 5.33 | 3.69 |
| NB | GA | 0.46 | 0.44 | 0.43 | 0.46 | 2.13 | 15.1 |
| NB | RFE | 0.49 | 0.49 | 0.49 | 0.49 | 1.19 | 2.98 |
| SVM | AFSA | 0.59 | 0.63 | 0.65 | 0.64 | 8 | 0.69 |
| SVM | All Features | 0.45 | 0.42 | 0.44 | 0.44 | 1 | 0.1 |
| SVM | FS | 0.61 | 0.58 | 0.67 | 0.6 | 2.46 | 6.05 |
| SVM | GA | 0.54 | 0.49 | 0.49 | 0.52 | 2.13 | 20.03 |
| SVM | RFE | 0.55 | 0.52 | 0.54 | 0.54 | 1.19 | 3.95 |

employing all features, suggests that the choice of feature selection method should be informed by considerations beyond mere accuracy metrics; factors such as computational efficiency, model interpretability, and domain-specific requirements emerge as critical determinants in selecting an appropriate feature selection strategy.

### D. COMPARATIVE ANALYSIS OF TRAINING TIMES

The training time efficiency of the Adaptive Feature Selection Algorithm (AFSA) has been analyzed in comparison to other prevalent feature selection methods across a variety of models and datasets. We aim to evaluate the extent of variability in training times and assess the statistical significance of these differences.

#### 1) VARIABILITY IN TRAINING TIMES

Upon comparing AFSA's training times with those of other algorithms, several key observations were made:

- **Maximum Difference in Training Time:** The most favorable scenario for AFSA showed it outpacing another method by approximately +8.71 seconds, indicating its potential for faster performance in optimal conditions.
- **Minimum Difference in Training Time:** Conversely, the most substantial lag observed for AFSA, compared to another method, amounted to -341.98 seconds, highlighting instances where AFSA's training time significantly exceeded that of its competitors.
- **Standard Deviation:** The calculated standard deviation of the training time differences between AFSA and other algorithms stood at 44.44 seconds, emphasizing the wide range of variability in training efficiency across different settings.

#### 2) STATISTICAL SIGNIFICANCE OF TRAINING TIME DIFFERENCES

A closer statistical examination of the training time differences between AFSA and each benchmarked algorithm yielded the following p-values:

- **All Features (no selection):** A p-value of 0.074 suggests a marginally significant difference, with AFSA generally exhibiting slower training times than when all features are used, albeit this being close to the conventional significance threshold.
- **Feature Selection (FS):** The observed p-value of 0.104 indicates a lack of statistically significant difference in training times between AFSA and FS.
- **Genetic Algorithm (GA):** With a p-value of 0.007, there is a statistically significant difference favoring AFSA's training time efficiency over GA.
- **Recursive Feature Elimination (RFE):** A p-value of 0.113 signals no statistically significant difference in training times when comparing AFSA to RFE.

The comprehensive analysis underscores the significant variation in training times when AFSA is juxtaposed with other feature selection methodologies. Notably, AFSA demonstrates a statistically significant efficiency advantage over the Genetic Algorithm, marking it as a potentially time-efficient choice under certain conditions. However, the broad standard deviation across comparisons signals that both dataset characteristics and model selection play pivotal roles in influencing training time outcomes. Consequently, in the selection of a feature selection method, considerations must extend beyond mere accuracy implications to include potential training time efficiencies or constraints, especially in contexts where time is of the essence.

### E. DISTINGUISHING FEATURE SELECTION IN MACHINE LEARNING AND DEEP LEARNING

In the realm of predictive modeling, it is essential to distinguish between the methodologies employed by traditional machine learning and deep learning frameworks, particularly concerning feature selection. Traditional machine learning models benefit substantially from explicit feature selection techniques, which aim to enhance model performance by reducing dimensionality, improving interpretability, and mitigating overfitting. In contrast, deep learning models, through their multi-layered architectures, inherently learn to identify and extract relevant features directly from raw data, bypassing the need for external feature selection processes. This inherent capability of deep learning models to autonomously learn complex representations makes them uniquely distinct from traditional approaches where feature selection plays a critical role. Our focus on refining feature selection for traditional machine learning models stems from this distinction, aiming to optimize model efficiency and performance in applications where explicit feature engineering is indispensable and where computational resources or data availability might limit the applicability of deep learning approaches.

### F. CHALLENGES AND POTENTIAL LIMITATIONS ASSOCIATED WITH THE PROPOSED ALGORITHM

The proposed Adaptive Feature Selection Algorithm (AFSA) is designed to significantly reduce the computational complexity of the feature selection process. However, this reduction can sometimes lead to lower overall evaluation results. A major challenge in this study was optimizing the thresholds and computational strategies to ensure both computational efficiency and superior evaluation outcomes.

The limitations of AFSA are rooted in its hybrid feature selection approach. While this approach is intended to be efficient, it may not fully capture all feature dependencies and interaction effects, which could compromise predictive accuracy. Additionally, although the adaptive threshold mechanism aims to enhance model accuracy and computational efficiency, the initial reliance on multiple ranking methods can lead to considerable computational overhead.

### G. SUMMARIZED INSIGHTS

Table 8 consolidates the performance metrics across all datasets, highlighting the relative effectiveness of different feature selection algorithms. FS emerges as the top-performing method, while AFSA demonstrates competitive performance, particularly in terms of recall and feature reduction factors. Table 9 summarises the performance metrics for different machine learning models, showcasing Decision Trees as the top-performing model. Notably, Naive Bayes exhibits efficient training times despite slightly lower overall performance. Table 10 offers insights into dataset-specific challenges and opportunities. The WOC2 dataset stands out with the highest average accuracy, while SSP

**TABLE 8.** Grouped by feature selection and summarised by the average of the evaluation metric for feature selection algorithms.

| Feature Selection | Accuracy | F1 Score | Precision | Recall | FRF | Training Time(s) |
|---|---|---|---|---|---|---|
| AFSA | 0.69 | 0.7 | 0.72 | 0.71 | 5.71 | 1.24 |
| All Features | 0.7 | 0.69 | 0.72 | 0.7 | 1 | 0.28 |
| FS | 0.76 | 0.75 | 0.77 | 0.76 | 1.8 | 14.6 |
| GA | 0.73 | 0.72 | 0.73 | 0.73 | 1.58 | 52.26 |
| RFE | 0.74 | 0.73 | 0.75 | 0.74 | 1.23 | 5.81 |

**TABLE 9.** Grouped by model and summarised by the average of the evaluation metric for models.

| Model | Accuracy | F1 Score | Precision | Recall | FRF | Training Time (s) |
|---|---|---|---|---|---|---|
| DT | 0.75 | 0.75 | 0.75 | 0.75 | 1.83 | 5.45 |
| KNN | 0.72 | 0.72 | 0.75 | 0.73 | 2.22 | 17.56 |
| LR | 0.71 | 0.71 | 0.72 | 0.71 | 2.53 | 10.21 |
| NB | 0.68 | 0.67 | 0.7 | 0.68 | 2.64 | 4.05 |
| SVM | 0.75 | 0.74 | 0.76 | 0.75 | 2.09 | 36.92 |

**TABLE 10.** Grouped by dataset and summarised by the average of the evaluation metric for datasets.

| Dataset | Accuracy | F1 Score | Precision | Recall | FRF | Training Time (s) |
|---|---|---|---|---|---|---|
| HESP | 0.52 | 0.52 | 0.55 | 0.53 | 3.49 | 6.69 |
| SSP | 0.66 | 0.66 | 0.7 | 0.67 | 2.07 | 35.89 |
| WOC2 | 0.94 | 0.94 | 0.94 | 0.94 | 1.32 | 8.8 |
| XAPI | 0.76 | 0.75 | 0.76 | 0.76 | 2.17 | 7.96 |

and XAPI demonstrate competitive performance with varying computational demands.

In conclusion, the experimental results underscore the efficacy of AFSA in feature selection for student performance prediction. While FS remains a top-performing method, AFSA provides a valuable alternative with significant feature reduction and efficient training times. The dataset-specific nuances highlight the importance of tailoring feature selection strategies to individual characteristics, offering valuable insights for practitioners in educational data analytics.

## VI. CONCLUSION & FUTURE WORK

In this study, we have introduced an Adaptive Feature Selection Algorithm (AFSA) for predicting student performance. The comprehensive evaluation, encompassing diverse machine learning models and student performance datasets, unveiled AFSA's superior performance compared to existing methods. The algorithm's prowess was evident, achieving an average feature reduction factor (FRF) of 5.71. This success underscores its efficacy in significantly reducing dimensionality while preserving predictive accuracy. AFSA emerges as a practical and promising tool for feature selection in educational data mining, with the potential to optimize learning paths and interventions for improved student outcomes. The results, as detailed in the Results and Discussion section (Section V), indicate that AFSA, while maintaining competitive accuracy, outperforms other feature selection algorithms in terms of feature reduction. The nuanced interplay between different feature selection methods, machine learning models, and datasets provides valuable insights for the educational data analytics domain. Notably, the forward selection (FS) algorithm consistently exhibited a lead in terms of accuracy, yet AFSA demonstrated comparable performance, emphasizing its reliability across various scenarios. The feature reduction factor achieved by AFSA surpassed alternative methods, emphasizing its potential to streamline datasets without compromising predictive accuracy. Moreover, AFSA showcased efficient training times, positioning it as a feasible choice for real-world applications.

Looking forward, there are several avenues for further research and refinement of AFSA, guided by the insights garnered from this study. Continued refinement of AFSA is warranted, exploring the integration of additional feature ranking methods and fine-tuning the iterative selection strategy. The exploration of hybrid approaches that leverage the strengths of different feature selection methods could enhance the algorithm's overall performance. Additionally, investigating the impact of hyper-parameter tuning on AFSA could uncover optimal configurations for specific datasets and models. This study revealed nuanced dependencies between feature selection techniques and machine learning models. Future research could delve deeper into understanding these interactions, potentially unveiling the most effective combinations for student performance prediction tasks. Investigating the impact of feature selection on emerging machine-learning models could contribute to the evolution of best practices in this dynamic field. While this study demonstrated AFSA's efficacy across diverse datasets, further exploration of its generalisability is essential. Testing the algorithm on larger and more diverse datasets, spanning various educational domains, will contribute to a comprehensive understanding of its robustness and applicability beyond the specific contexts explored in this study. Exploring the relationships between the features selected by AFSA and student outcomes holds promise for developing targeted interventions and support systems. Understanding the underlying factors contributing to student success or challenges can inform the design of personalized learning strategies and interventions, contributing to more effective educational practices. In conclusion, our study establishes AFSA as a cutting-edge feature selection algorithm with significant potential for enhancing student performance prediction in educational data mining.

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